

Direct-to-Consumer Eyewear Funnel Analysis (SQL + Python)

Product Funnel, A/B Testing, and Preference-Based Conversion

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NOTE: This project analyzes user behavior across Warby Parker's onboarding and purchase funnels using SQL and Python. The focus is on understanding where users drop off, how product design choices affect conversion, and what insights can be drawn from preferences shown during onboarding.

All data used in this project is fictional and provided as part of a Code Academy case study developed in collaboration with Warby Parker's Data Science team.

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1. Executive Summary

1.1 Key Questions and Metrics

Key Data Questions

- What can be inferred about the drop-off distribution of users across all stages of onboarding?
- A/B Test: Does sending 5 pairs vs 3 pairs (equal sample size 50/50) increase conversion rate?
- What are some key insights from conversion rates?

Success Metrics

- Conversion rates from SQL funnels and Python automation

1.2 Key Findings

- Significant survey attrition recorded between early preference questions and more cognitively-involved questions
- Marginally higher drop-off rates observed between "Home Try-On" and "Purchase" stages of the funnel
- A/B test show that customers receiving 5 pairs for "Home Try-On" are significantly more likely to convert
- Steps causing increased cognitive friction show higher drop-off rates

1.3 Recommendation Snapshot

- Simplify or defer more involved steps
- Favor the 5-pair “Home Try-On” variant or dynamically assign it to high-intent users
- Use early style signals to tailor messaging and product exposure

2. Data Sources and Pipeline

2.1 Tables

1. `survey` (event-level) / `quiz` (user-level snapshot)

- Users are asked 5 questions regarding product attributes (style, fit, shapes, colors) and last eye exam
- Unique `User_ID` numbers are recorded for each entry in the survey table
- The `quiz` table stores final selections per user for analysis

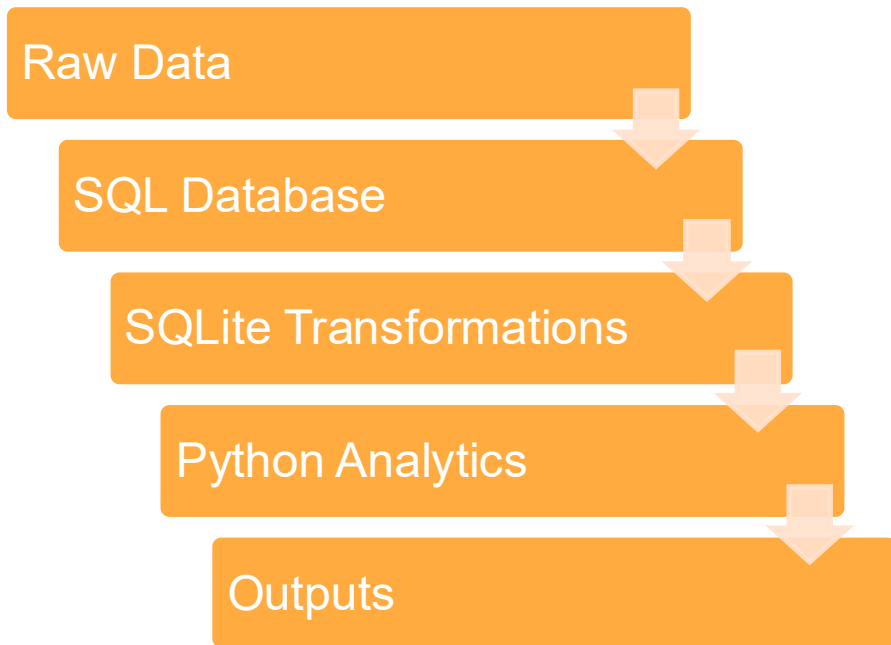
2. `home_try_on`

- A/B test conducted where customers are split into two groups based on `number_of_pairs`
- `number_of_pairs` = experiment variant (3 vs 5)
- `address` captured for shipping/logistics

3. `Purchase`

- One row per purchase (`user_id` + product attributes + `price`)
- Used for conversion lift analysis

2.2 Data Pipeline and Design



- Raw data ingested locally via `build_db_wp.py`
- Data normalized into a relational SQL database `warby_parker.db`
- Data visualized prior to analysis in `queries.sqlite`
- Core analytics logic implemented in SQL via `pandas` library's function `read_sql()` and the database driver `sqlite3`
- User-level funnel constructed via `LEFT JOINS` from `quiz`, `try_on`, and `purchase` tables
- Survey responses aggregated to measure per-question drop-off rates
- A/B test segmented by `number_of_pairs` to compute the conversion lift (50% 3-pairs, 50% 5-pairs)

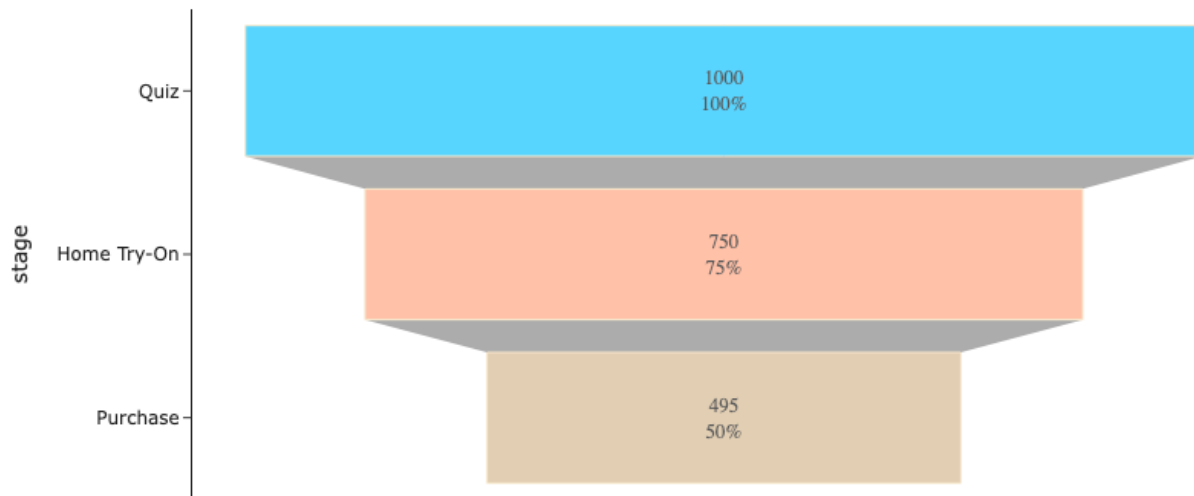
2.3 GitHub Repo

```
eyeware_funnel_analysis/  
├── outputs/  
│   ├── figures/  
│   │   ├── conversion_funnel.png  
│   │   ├── survey_completion_funnel.png  
│   │   └── ab_test_purchase_rate.png  
├── conversion_by_style_preference.png  
│   └── tables/  
│       ├── conversion_rates.csv  
│       ├── ab_test_metrics.csv  
│       └── style_conversion_rates.csv  
├── warby_parker.db  
├── queries.sqlite  
├── analysis_funnel.py  
├── analysis_ab_test.py  
├── analysis_demographic.py  
└── wp_deliverable.pdf
```

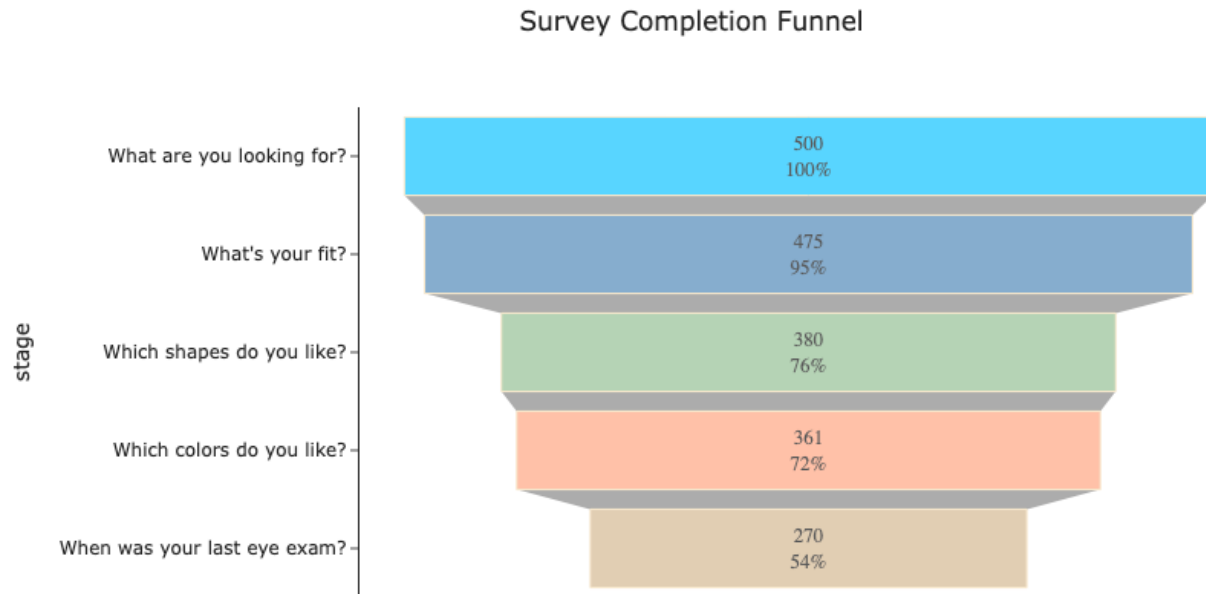
3. User Funnels

3.1 User Conversion Funnel between Stages of the Entire Conversion Process

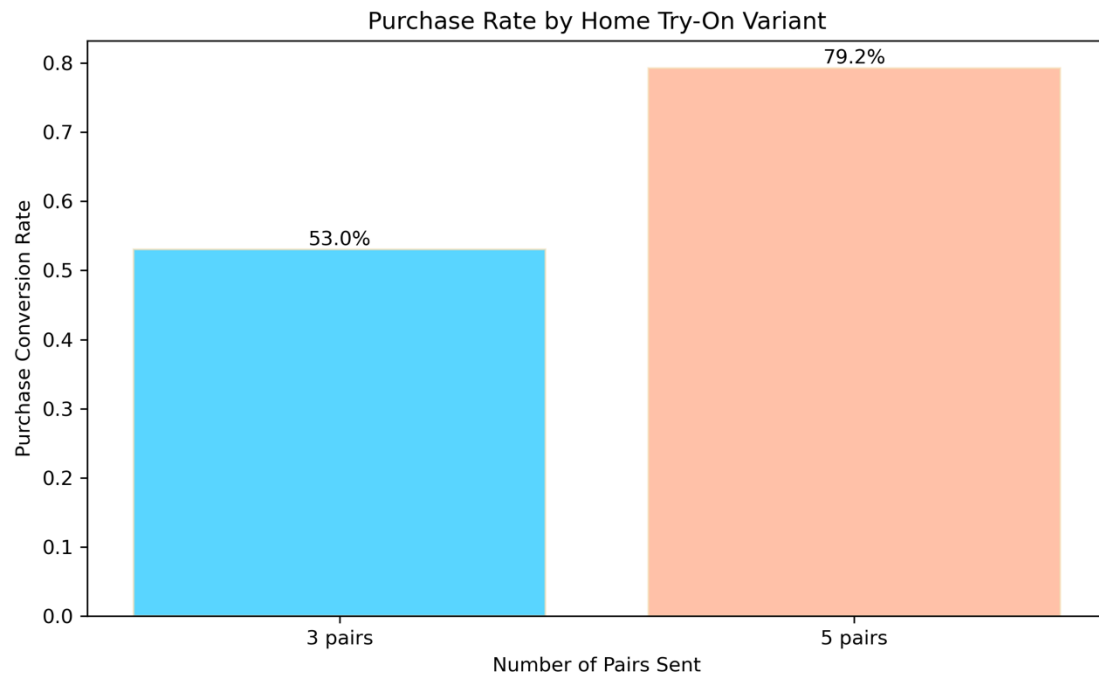
Warby Parker User Conversion Funnel



3.2 User Conversion Funnel during the Initial Stage of the Conversion Process



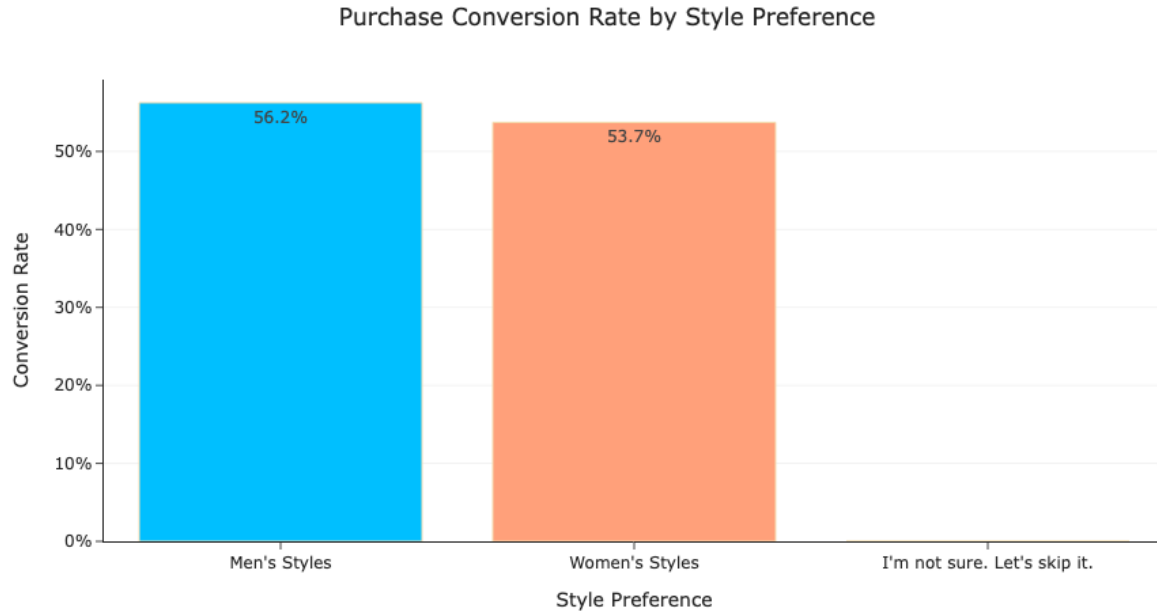
3.3 Home Try-On A/B Test



A/B Test Split

- **Control:** 3-pair Home-Try-On (50% of the users participating in the program)
- **Variant:** 5-pair Home-Try-On (50% of the users participating in the program)

3.4 Conversion Rates by Style Preference



4. Summary and Limitations

4.1 Summary

- Identified substantial drop-off across both Stage 1 (quiz to home-try-on) and Stage 2 (home-try-on to purchase) of the funnel
- Stage 2 exhibits a marginally higher drop-off rate
- Within the quiz, the largest drop-offs occur at higher-effort questions related to fit and eye exam history (~19% and ~18% user loss respectively)
- The Home Try-on A/B test shows that customers who received 5 pairs converted at a rate of 26.2% higher than the 3-pair control group
- Conversion rates between men's and women's styles show only marginal differences, with men having approx. 2.5% higher conversion rates than women

4.2 Limitations

The following is NOT an extensive list of limitations one must be aware of when referring to this analysis:

- A/B test analysis assumes random assignment and does not account for potential confounding variables (e.g., prior intent, prescription readiness)
- Conversion outcomes are measured at the user-level and do not incorporate repeat visits or delayed purchases
- Funnel analysis is observational; causal drivers of drop-off cannot be fully isolated
- Other variables such as promotions and temporal effects (seasonality, delayed purchases, etc.) are not examined

5. Next Steps

5.1 Next Steps

- Account for delayed purchases by tracking users across sessions using a consistent identifier (e.g., hashed IP or device-based ID)
- Incorporate promotions and discount codes to control for price-driven conversion effects
- Reduce quiz frictions by simplifying higher-effort questions (e.g. use an approximate time from the last eye exam instead of exam date)
- Expand the 5-pair Home Try-On option to validate the results in a longer-running experiment environment
- Evaluate downstream metrics such as average order value and return rates to complement conversion

